INCH-POUND

MIL-E-85082/4

16 November 1992

MILITARY SPECIFICATION SHEET

ENCODER, SHAFT POSITION TO DIGITAL
CONTACT TYPE, ABSOLUTE
TYPE 11KCL13R08A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Encoders, Shaft Angle to Digital, described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-E-85082.

TABLE I. Requirements.

Requirements	Value	Unit	Tolerance
Code and scan	Binary, V-Scan		
Capacity	2^{13}	counts	maximum
Resolution	2^{8}	counts/turn	
Full count revolutions	32	turns	
Voltage interrogation	15	volts DC	maximum
Contact current	3.0	mA/bit	maximum
Transition accuracy	±1/2	bit	maximum
Starting torque	0.5	oz-in	maximum
Running torque	0.4	oz-in	maximum
Operating speed	200	rpm	maximum
Slew speed	1000	rpm	maximum
Radial play	0.0010	inches	maximum
End play	0.0020	inches	maximum
Total shaft runout	0.0010	inches	maximum
Concentricity	0.0010	inches	maximum
Perpendicularity	0.0010	inches	maximum
Operating temperature	-55° to +85°	degrees C	min-max
Moment of inertia	5.0	$gm\text{-}cm^2$	maximum
Weight	4.0	oz	maximum
Life	10,000,000	revolutions	minimum

AMSC - N/A FSC 5990

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TABLE II. Part identifying number variant characteristics.

Part Identifying No. 1/	A ±0.015	E +0.000 -0.001	G Maximum	Shaft Style	Terminal End	U +0.0000 -0.0001
M85082/4-01A	.500	.440	1.700	Plain	Wire Leads	.1248

1/ Part identifying number M85082/4-01A shall be the NATO standard.

1/ The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE III. Wire termination identification.

DIGIT	LAG	LEAD
2^0	WHITE/BLUE/BLACK	
2^1	BROWN	WHITE/BROWN
2^2	RED	WHITE/RED
2^3	ORANGE	WHITE/ORANGE
2^4	YELLOW	WHITE/YELLOW
2^5	GREEN	WHITE/GREEN
2^6	BLUE	WHITE/BLUE
2^7	VIOLET	WHITE/VIOLET
2^8	GRAY	WHITE/GRAY
2^9	WHITE	WHITE/BLACK
2^{10}	ORANGE/BLACK	WHITE/ORANGE/BLACK
2^{11}	YELLOW/BROWN	WHITE/YELLOW/BROWN
2^{12}	BROWN/RED	WHITE/BROWN/RED
	COMMON	BLACK

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NOTES:

1. Radial play and end play shall be measured with a reversing load of 0.4 oz maximum applied 0.12 inch from end of shaft.
2. Dimension H shall be 0.030 with a tolerance of ±0.010.
3. Dimension I shall be 0.062 with a tolerance of +0.000/-0.005.
4. Dimension J shall be 0.090 with a tolerance of +0.005/-0.000.
5. Mechanical shock shall be conducted at peak of 15 g's for a duration of 11 ms.
6. Vibration shall be conducted at peak of 15 g's with a frequency range of 5 to 500 Hz.
7. Altitude tests shall be performed at a height of 70,000 feet maximum.
8. Rotation for increasing count shall be clockwise.

Custodians:
Navy - AS
Army - AR
Air Force - 85

Review activities:
DLA - ES

Preparing activity:
Navy - AS

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